Rolling Shutter Design Calculation

Given Data.:

Shutter Width.:12700 mm

Shutter Height.: 2750 mm

Wind Load.: 14km/hr

GI Slat Weight: 650 kg

Intermediate Support tube : 432 kg

A) <u>For Deflection Calculation</u>.:

1.Wind Speed.:

$V = 14{,}000/3600$

$= 3.89$ m/sec.

2.Wind Pressure (P).:

$\mathbf{P} = 0.613 \times V^2$

$= 0.613 \times (3.89)^2$

$= 0.613 \times 15.1321$

Wind Pressure (P) = 9.275 N/m^2 or Pa

3.Convert Given Load.:

Given load = GI Slat Weight + Shaft weight

$= 650 + 432$

Total Weight = 1082 kg per meter

$= 1082/12.7 = 81.25$ kg per meter

Convert to 'N/m' –

= Mass x $Accl^n$ due to gravity

$= 81.25 \times 9.81$

$= 797.15$ N/m

Convert to 'N/mm' –

$= 797.15 /1000$

Total Given Load = 0.797 N/mm

Wind load per meter 'Width'

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Area per meter Width = 12.7x2.750

= 34.95m^2

4.Wind Load (Wind force per meter).:

= P x A

= 9.275 x 34.95

= 323.92 N/m

Convert to 'N/mm' -

= 323.92/1000

= 0.323 N/mm

Where:

P = 9.275 N/m^2

A = Width X Height

= 12.7 X 2.750

A = 34.92 m^2

5.Total Weight per 'mm'

Total Weight = Total Given Load + Wind load per meter

= 0.797 + 0.323

= 1.12 N/mm

6.Section Properties.:

OD = 210 mm

ID = 198 mm

Thickness (t) = 6 mm

d = D-2t

=210-12

d = 198 mm

7.Movement of Inertia (I).:

$I = (\pi (D^4 - d^4))/ 64$

$I = ((3.14*(210^4 - 198^4)) / 64$

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$I = 20010453.84\ mm^4$

8.Section Modulus (Z).:

$Z = I / Y$

Where:

$Y = D/2$

$= 60/2 = 105$ mm

$Z = 20010453.84 / 105$

$Z = 190575.75\ mm^3$

Convert shutter weight to force

$W = 650x9.81$

$=6376.5$ N

Assume uniform load on shaft

$ɯ=6376.5/13700$

$ɯ = 0.465$ N/mm

9.Deflection Calculation.:

Simply Supported Beam.:

$= 5\ ɯ\ L^4 / 384EI$

$= (5 \times 0.465 \times 12900^4) / (384 \times 2x10^5 \times 20010453.84)$

Deflection =41.89 mm

10.Allowable deflection

For Roof:

Allowable = L/400

= 12900/400

= 32.25 mm

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10.Final Result.:

Calculated deflection = 41.89mm

Allowable deflection = 32.25mm

The shaft will sag about 42 mm

Actual site shaft deflection =55 mm

B) <u>For Bending Stress Calculation</u>

1.Maximum Bending Movement.:

$M = Ш L^2/8$

$= (0.465 \times 12900^2)/8$

$= 9672581.25$ N-mm.

2.Section Modulus (Z) = 190575.75 mm^3

3.**Bending Stress ('N/mm').:**

Bending Stress = M/Z

= 9672581.25 / 190575.75

= 50.75 N/mm^2

= 50.75 M pa.

Allowable Stress = 0.5* fy

Or

Allowable Stress = fy/Factor of Safety

For MS-Yield Strength = 250 MPA & Ultimate Strength = 400 MPA

1) Allowable stress = 0.5 x 400
   = 125 N/mm^2 allowable stress

4.Final Result.:

50.75 M pa is less than 125 M pa allowable stress

Since**,**

Design is Safe in Bending Stress.